Title Page

1. Introduction & Problem Description
2. Proposed Solution
3. Class Diagram
4. GitHub Repository

Smart Docklands Parking Management System

Data Structures & Algorithms

Continuous Assessment Project

Student Name: Aaron Bridger

Student Number: 24247421

National College of Ireland
BSc (Hons) in Computing

Submission Date: 15 March 2026

Section 1 – Introduction & Problem Descriptions:

For the Smart Docklands I'll be trying to initiative aims to try encouraging innovation and technological solutions to try improving the efficiency, sustainability and the quality of life for the people within the Dublin Dockland Area. As the Docklands has developed into a big business and technology hub, it has experienced a significant increase in traffic and vehicular usage. Probably the one of the biggest issues is to be faced in urban areas such as the Docklands. Now, that's why for the management of the parking spaces it will be crucial part of the project. The drivers will be more and likely to spend most of their time looking for a opened parking spot, which can also help to contribute to congestion and increased fuel usage.

The parking management system will always play a big role in trying to improve the urban mobility for people. We will do this by organising parking spaces more efficiently and tracking vehicle usage every day, such systems can help with reducing the time drivers spend searching for available spots. Also, implementing a digital system that will manage people's vehicles and parking locations can provide an easy and clear structured way to track parking availability and improve overall parking for the future of Docklands.

My project will propose the development of a Smart Docklands Parking Management System applied in Java using a Swing graphical user interface. The application will be designed to show how parking spaces and vehicles can be sorted and managed using data structures and object-oriented programming principles. For the main goal of the system, I'll try to demonstrate how structured data management can improve the organisation of parking operations within a smart city environment such as Dublin Docklands.

Section 2 – Proposed Solution:

The proposed application is a Smart Docklands Parking Management System that will allow the users to be able to manage parking spaces and vehicles through a Java Swing interface. Also, the system will model the vehicles, parking spaces and parking management operations using object-oriented design. The application will also allow vehicles to be added to the system, placed into a waiting queue, and assigned to available parking spaces when they become free. The system also tracks recently parked vehicles and retains a structured list of all parking spaces.

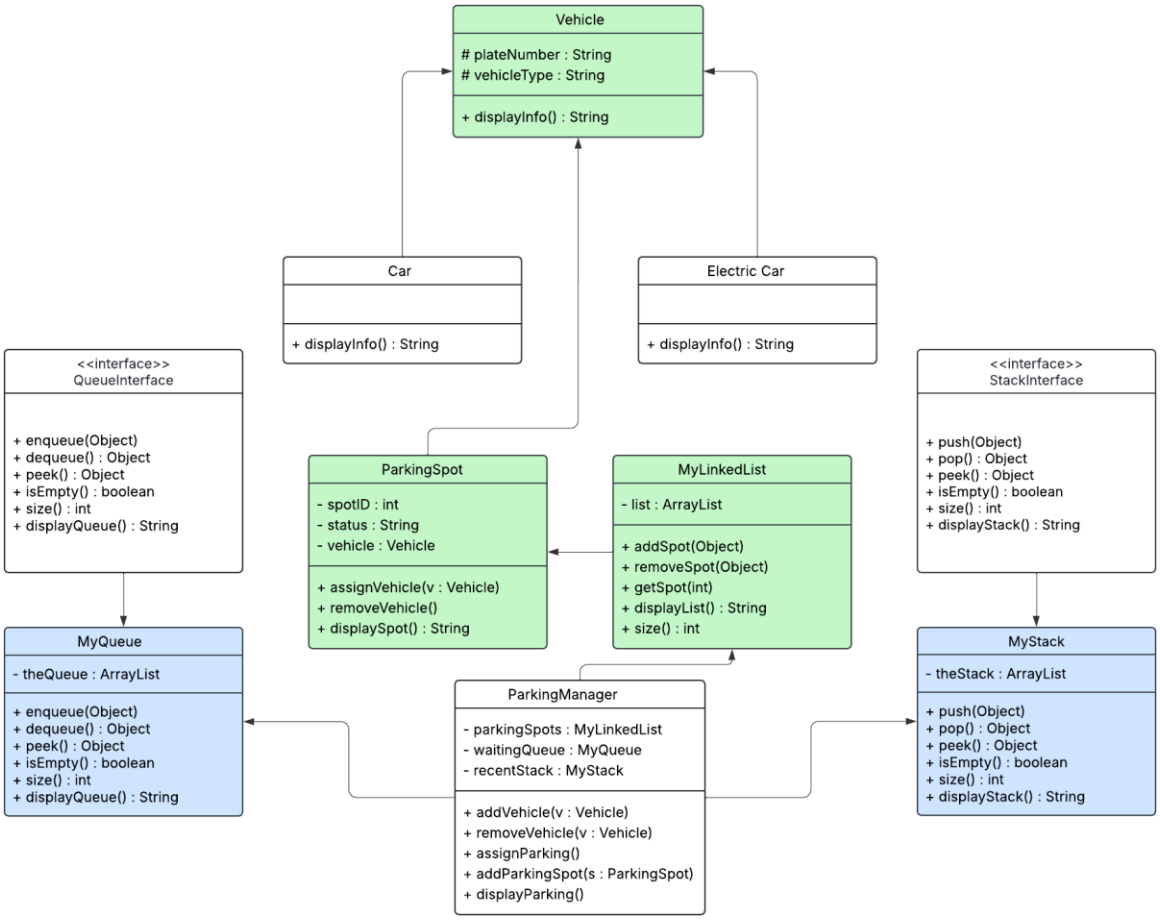
For my project the system will be designed using several core classes including Vehicle, Car, Electric Car, Parking Spot and Parking Manager. The class called Vehicle will act as a parent class that defines the basic properties of all the vehicles such as the plate number and vehicle type. Two subclasses, Car and Electric Car, inherit from this class and will help to show the belief of inheritance and polymorphism in object-oriented programming.

To try properly managing the parking data, the system will have to be using three Abstract Data Types, a stack, a queue and a singly linked list. The queue is used to help with managing vehicles that are waiting for parking spaces, this will be ensuring that vehicles are assigned parking in a who comes first will be first served. Also, the stack is used to track recently parked vehicles which will be allowing the system to easily display the most recent parking activity. The singly linked list will store all available parking spots and allows parking spaces to be added, removed or accessed properly.

The interfaces are going to be used to define the behaviour of the stack and queue data structures. The Stack Interface and Queue Interface specify the operations that must be applied by their corresponding classes, My Stack and My Queue. This approach demonstrates abstraction and helps to promote a modular design structure for the Docklands.

So, overall, the system will be showing how data structures and object-oriented programming techniques can be used to arrange and manage parking data efficiently. The application reflects the goals of the Smart Docklands program by presenting a technology-based approach to improving the urban parking management for people.

Section 3 – Class Diagram:



Explanation Of the Diagram:

For the class diagram that I pasted above. Is for me to illustrate the overall structure of the Smart Docklands Parking Management System. The system is going be designed with object-oriented programming principles such as inheritance, interfaces and abstraction. For the Vehicle class acts as the parent class for the Car and Electric Car classes, showcasing inheritance and polymorphism. The application will be also using three abstract data types to manage parking data professionally. A queue will also be used to manage vehicles that are waiting for a parking space, a stack is used to track recently parked vehicles in them spaces, and a singly linked list is used to store parking spots. The interfaces are used to define the behaviour of the stack and queue functioning, improving modularity and maintainability of the system design. The Parking Manager class acts as the main controller that helps to manage all the parking spaces, vehicles and data structures within the system for Docklands.

Section 4 – GitHub Repository:

Now, for the source code of my project it's going to be held inside a public GitHub repository so it can be viewed publicly. The repository will have the NetBeans project files as well as the project report. For the development process it'll be documented through a series of commits. Such as 9 different commits will be demonstrating the execution of the different components of the system, which will be including the interfaces, abstract data types, application classes and graphical user interface. The GitHub repository will be showing clear evidence of the development process throughout the 9 commits and the version control throughout the project.

Section 5 – Implementation Overview

The Smart Docklands Parking Management System was built using Java with a NetBeans Swing GUI. It allows the users to manage parking spaces and vehicle details easily. The ParkingManager class checks available spots using a linked list, assigns vehicles if a free space exists, and logs them in a stack for recent activity. If full, well then, the vehicles join a queue until a spot opens, ensuring first-come, first-served processing. The GUI connects to the backend, updates in real time, and displays parking status, demonstrating efficient use of data structures and object-oriented programming in smart city applications.

GITHUB LINK:

https://github.com/AaronBridgerNci/SmartDocklandsParkingSystem.git